JMR3E.qxdII 7/3/03 9:51 AM Page 310

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This article reports the development and validation of a parsimonious, generalizable scale that measures the hedonic and utilitarian dimensions of consumer attitudes toward product categories and different brands within categories. The hedonic/utilitarian (HED/UT) scale includes ten semantic differential response items, five of which refer to the hedonic dimension and five of which refer to the utilitarian dimension of consumer attitudes. The authors conducted six studies to establish the unidimensionality, reliability, and validity of the two HED/UT subscales. In reaching the final scale, the authors also develop and implement a unique process of paring down a psychometrically sound but otherwise too large set of items. Nomological validity is established by replacing a typical, one-dimensional attitude toward the brand measure with the hedonic and utilitarian dimensions in a central route processing model. Results suggest that the hedonic and utilitarian constructs are two distinct dimensions of brand attitude and are reliably and validly measured by the HED/UT scale.

Measuring the Hedonic and Utilitarian Dimensions of Consumer Attitude

Researchers continually seek a richer understanding of consumer attitudes. Investigation of the hedonic and utilitarian components of attitude has been suggested in such diverse disciplines as sociology, psychology, and economics. This multidisciplinary recognition of the hedonic and utilitarian elements of consumption mirrors parallel theoretical development in marketing, mainly from a series of articles by Hirschman and Holbrook (1982; Holbrook and Hirschman 1982). Although product/brand attitude has historically been treated one-dimensionally (e.g., Osgood, Suci, and Tannenbaum 1957), the view that attitudes are complex and multidimensional (e.g., Bagozzi and Burnkrant 1979; Eagly and Chaiken 1993) has directed marketing researchers toward integrating an experiential view of consumption with more-traditional functional approaches (e.g., Mano and Oliver 1993; Park, Jaworski, and MacInnis 1986). In an early attempt to measure multiple dimensions of product/brand attitudes, Batra and Ahtola (1990, p. 159, emphasis added) state, "consumers purchase goods and services and perform consumption behaviors for *two* basic reasons: (1) consummatory affective (hedonic) gratification (from sensory attributes), and (2) instrumental, utilitarian reasons." We adopt this two-dimensional conceptualization of consumer attitudes: The first dimension is a *hedonic* dimension resulting from *sensations* derived from the experience of using products, and the second is a *utilitarian* dimension derived from *functions* performed by products.

Measurement of these attitudinal dimensions can provide researchers and managers with fresh approaches to modeling marketing problems. Measures of attitudinal dimensions provide building blocks for researchers attempting to develop models that explain a greater proportion of the variance in consumer behavior (Bagozzi and Burnkrant 1979; Olney, Holbrook, and Batra 1991). Measures of the hedonic and utilitarian dimensions of attitude enable marketers to test the effectiveness of advertising campaigns that stress experiential or functional positioning strategies (Park, Jaworski, and MacInnis 1986). Then, too, these measures can reveal brand differences/positions that may not be apparent when a single dimension attitude measure is used (Dillon et al. 2001; Machleit, Allen, and Madden 1993). Previous research has suggested that products/brands that are highly valued on the hedonic dimension rather than the utilitarian dimension are better able to charge a price premium (Dhar and Wertenbroch 2000) or engage in sales promotions (Chandon, Wansink, and Laurent 2000). Thus, measures of these two dimensions may serve as input into pricing and sales promotion decisions. Relatedly, because

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JMR3E.qxdII 7/3/03 9:51 AM Page 311

owners of hedonic items keep them for longer periods (Dhar and Wertenbroch 2000) and because ownership is a recommended segmentation criterion (Kirmani, Sood, and Bridges 1999), measuring attitude dimensions is a basis for segmenting markets.

A major impediment to such research is the difficulty of capturing both dimensions with a reliable and valid measurement instrument. The most commonly used scale, which Batra and Ahtola (1990) developed, has proved problematic in nearly all published reports of its use (e.g., Chaudhuri and Holbrook 2001; Crowley, Spangenberg, and Hughes 1992). This performance can be explained both theoretically and methodologically. A prominent criticism of the scale is its inability to account for relevant theoretical concepts within a nomological framework. For example, the instrument does not accommodate involvement, as Hirschman and Holbrook (1982) propose. Predictably, Batra and Ahtola's items cross-load with items from Zaichkowsky's (1985) measure of product category involvement, suggesting inadequate discriminant validity (Mano and Oliver 1993).[1] This result in part can be accounted for by an apparent concept-X scale interaction suggested by the inconsistent factor loadings of scale items reported by the original authors. A careful reading suggests that Batra and Ahtola blur the theoretical distinction between *unidimensionality* (Gerbing and Anderson 1988) and *single-dimension scales*.[2] In developing multiple item measures, theory suggests that researchers should sample from the (hypothetical) set of all items representing the construct of interest (Churchill 1979). However, Batra and Ahtola conduct two independent development efforts with different sets of initial items that ostensibly represent the same construct(s) and subsequently combine items from the two efforts to arrive at a final scale. Because slight differences in initial construct definition or domain sampling can generate items from different domains, this method can introduce concept-X scale interaction, which could easily explain the item cross-loading problem Mano and Oliver (1993) identify. Batra and Ahtola do not report a test of unidimensionality for their measure; thus, the possibility exists that the items actually represent different but correlated domains.

The measure has also disappointed with regard to predictive validity and generalizability across product categories (Crowley, Spangenberg, and Hughes 1992). Although Batra and Ahtola (1990) report initial, successful applications of their scale in terms of evaluation of specific branded products, Crowley, Spangenberg, and Hughes (1992) are unable to replicate the original findings or demonstrate generalizability of the scale to product categories. Product categories in the Crowley study scored high on both dimensions of Batra and Ahtola's scale, inappropriately appearing in the same quadrant when represented in a two-dimensional space. The study's results arguably are related to the methods used in the original validation studies. Batra and Ahtola's tests of validity compare their apparent hedonic and utilitarian dimensions of consumer attitudes with each other or with consumer evaluations of product attribute levels. Although it is not methodologically flawed, such a procedure is theoretically inadequate for establishing the validity of a scale purporting to measure separate dimensions of consumer attitudes toward product categories and/or brands in a given category. A scale must be tested within a nomological network that shows its relationship or lack of relationship to other constructs. As a disquieting example, in one series of factor analyses, Batra and Ahtola find that their "hedonic subdimension" loads on the same factor as items purporting to measure overall brand attitude, which indicts the discriminant validity between these two measures. Furthermore, Batra and Ahtola do not present evidence linking their scales to a higher-order attitude construct. Although we embrace the two-dimensional view of consumer attitudes, Batra and Ahtola's results—together with the findings of Crowley, Spangenberg, and Hughes; Mano and Oliver (1993); and others—call into question their instrument's reliability and validity.

Our research contributes to the integration of the experiential view of consumption in marketing research by developing a rigorously tested, reliable, valid, and generalizable scale to measure the hedonic and utilitarian dimensions of consumer attitudes. Following accepted scale development procedures (e.g., Churchill 1979; Gerbing and Anderson 1988), we began with a pool of items that represented the domains of interest and winnowed this pool down to ten items that displayed sound psychometric properties. In a series of studies, we demonstrate the scale's appropriateness in tests of unidimensionality and reliability as well as discriminant, predictive, and nomological validity. We demonstrate predictive validity by plotting scale scores for both product categories and brands in a two-dimensional space; for some brands/products the hedonic dimension may be more important, for others the utilitarian dimension may prevail, and for still others both dimensions may be important (Batra and Ahtola 1990; Kempf 1999). Finally, we placed our scale within a nomological network by proposing and testing a central route processing model that compares the hedonic and utilitarian dimensions to brand attitude. Notably, we present an empirical method we developed for reducing a qualified but lengthy scale to a parsimonious set of items. We describe our scale development process in detail in the following section.

SCALE DEVELOPMENT

Study 1: Initial Item Generation and Selection

We derived an initial pool of adjective pairs from published research and a pretest of students and professionals. We gave participants definitions (Hirschman and Holbrook 1982) of both hedonic and utilitarian (HED/UT) scale dimensions (but not dimension labels) and asked them to list adjectives that would capture these definitions and to provide the polar opposite adjective. We eliminated duplicates or items found psychometrically inadequate in published literature. We asked half of 608 student subjects from a large

[1]Several definitions exist for discriminant validity. Some researchers define the term as demonstrating that a measure does not correlate with theoretically unrelated constructs (e.g., Tull and Hawkins 1993). Others define the term as too-high correlations with other tests from which they were intended to differ (e.g., Bagozzi, Phillips, and Yi 1991; Nunnally and Bernstein 1994). The second definition is the one we adopt in this study.

[2]Unidimensionality necessitates that all items in a scale are generated from the same conceptual domain (Gerbing and Anderson 1988). However, it is possible for items appearing methodologically united on a single dimension to derive from theoretically different but correlated domains, that is, concept-X scale interaction. A concept-X scale interaction occurs when measure items are affected by a construct other than the hypothesized construct.

JMR3E.qxdII 7/3/03 9:51 AM Page 312

North American university to rate one of the following brands on the resultant 27, seven-point semantic differential items: IBM personal computers, Lay's potato chips, Hilton vacation resorts, Wesson cooking oil, Dawn dish detergent, and Nike athletic shoes. We asked the other subjects to rate one product category associated with each brand, with no mention of the brand. We selected categories and brand names from a pretest of 70 students who were asked to (1) "List products or services purchased predominantly for pleasure/utility," and (2) "Within that category [referring to (1)], what brand first comes to mind?"

Psychometric analysis. We conducted principal components exploratory factor analysis (EFA) separately for both brands and product categories, and we assessed internal consistency and item-to-total correlations separately (Churchill 1979). All items loaded as predicted, save for three items dropped because of low item-to-total correlations and factor loadings; deleted pairs were "hedonic/not hedonic," "utilitarian/not utilitarian," and "convenient/not convenient." Thus, EFA and internal consistency estimates suggested a two-dimensional scale (HED/UT) with 12 adjective pairs representing each of the hedonic and utilitarian dimensions of product attitude. Table 1 contains items, factor loadings, item-to-total correlations, reliability, and average variance extracted (AVE) estimates (Fornell and Larcker 1981).

We assessed unidimensionality using confirmatory factor analysis (CFA) in LISREL 8 (Jöreskog and Sörbom 1996). The CFA indicated that for both brand names and product categories, the items loaded as predicted and with minimal cross-loading, providing evidence of unidimensionality

Table 1
HED/UT ITEMS: INITIAL AND FINAL SCALE STATISTICS

	Study 1[b]						*Study 2*[c]
	Brand Names			*Product Categories*			
Items[a]	*Factor 1*	*Factor 2*	*Item-Total Correlation*	*Factor 1*	*Factor 2*	*Item-Total Correlation*	*Product Categories*
Utilitarian							
Effective/ineffective	.68	.58	.84	.60	.62	.86	.87/.84
Helpful/unhelpful	.66	.59	.80	.58	.61	.84	.89/.86
Functional/not functional	.57	.64	.77	.63	.54	.82	.91/.87
Necessary/unnecessary	.69	.43	.73	.49	.65	.75	.89/.86
Practical/impractical	.58	.54	.75	.51	.63	.75	.89/.86
Beneficial/harmful	.69	.46	.75	.47	.68	.76	
Useful/useless	.74	.34	.73	.46	.66	.73	
Sensible/not sensible	.66	.50	.74	.52	.61	.78	
Efficient/inefficient	.69	.47	.74	.49	.63	.78	
Unproductice/productive	.56	.46	.75	.53	.55	.70	
Handy/not handy	.53	.46	.59	.47	.43	.66	
Problem solving/not problem solving	.44	.56	.57	.61	.56	.66	
Hedonic							
Not fun/fun	–.62	.56	.82	.72	–.48	.84	.98/.85
Dull/exciting	–.70	.46	.72	.61	–.57	.80	.85/.80
Not delightful/delightful	–.61	.56	.79	.69	–.46	.81	.86/.81
Not thrilling/thrilling	–.67	.56	.84	.66	–.51	.80	.82/.77
Enjoyable/unenjoyable	–.62	.45	.71	.63	–.44	.72	.79/.74
Not happy/happy	–.57	.66	.83	.73	–.42	.80	
Unpleasant/pleasant	–.66	.45	.74	.61	–.51	.76	
Not playful/playful	–.58	.64	.82	.62	–.52	.76	
Cheerful/not cheerful	–.60	.51	.75	.69	–.43	.76	
Amusing/not amusing	–.57	.40	.64	.61	–.46	.71	
Not sensuous/sensuous	–.47	.57	.68	.60	–.44	.70	
Not funny/funny	–.39	.52	.60	.63	–.32	.65	
Explained Variance[b]	33%	60%		32%	63%		80%
AVE[d]							
Hedonic	50%			47%			71%
Utilitarian	49%			49%			79%
Reliability[d]							
Hedonic	.92			.91			.95
Utilitarian	.92			.92			.93
Coefficient Alpha							
Hedonic	.95			.95			.95
Utilitarian	.95			.95			.92

[a]Items in bold compose the final HED/UT scale.
[b]From principal components factor analysis (unrotated solution).
[c]From CFA (factor loadings/item-total correlations).
[d]Calculations described by Fornell and Larcker (1981).

JMR3E.qxdII 7/3/03 9:51 AM Page 313

(Gerbing and Anderson 1988). For the two-factor brand name model, χ^2 = 901.77 (degrees of freedom [d.f.] = 251, $p < .001$); goodness-of-fit index (GFI) = .80; normed fit index (NFI) = .86; nonnormed fit index (NNFI) = .88; and comparative fit index (CFI) = .90.[3] For the two-factor product category model, χ^2 = 1062.16 (d.f. = 251, $p < .001$); GFI = .77; NFI = .84; NNFI = .87; and CFI = .88. Model fit represented a significant improvement over the null model for both brands (χ^2 = 6110.2 [d.f. = 276]) and product categories (χ^2 = 5777.0 [d.f. = 276]).

Study 2A: Scale Reduction

Although the previous development procedure followed published recommendations, CFA fit statistics suggested that further improvement in the scale was possible. High interitem correlations for the two subscales suggested that items could be dropped while maintaining reliability. Moreover, this scale was too long to be pragmatic for inclusion in most research. Thus, we reexamined the scale with a new group of 400 students at the same university. Subjects rated two randomly assigned product categories (selected from the studies of Batra and Ahtola [1990]; Crowley, Spangenberg, and Hughes [1992]; and Spangenberg, Voss, and Crowley [1997]) on the 24 HED/UT items.[4] After elimination of nonrespondents, 706 completed product category evaluations were available for analysis.

In Study 1, scale development depended on the evaluation of item-to-total correlations, internal consistency (reliability), AVE, and unidimensionality. Although the literature typically advocates scale purification through elimination of items with regard to these characteristics (see DeVellis 1991; Spector 1992), no criteria are published regarding paring down a psychometrically acceptable but otherwise too large set of items. Thus, to reduce scale length, we created a procedure that uses CFA fit indexes and χ^2 difference testing. Conducting separate analyses for each subscale, we selected and eliminated the scale item with the lowest item-to-total correlation. We then conducted a χ^2 difference test between the original CFA model and a CFA model of the reduced scale. In addition to difference testing, we also noted GFIs and AGFIs at each step. The GFI will increase as items are removed; however, AGFI could decrease. If the difference test provided a significant result and AGFI increased, we accepted the reduced scale as the better of the two. We iterated the process by recomputing item-to-total correlations, deleting the lowest item, fitting a CFA model, and conducting the χ^2 difference test. We halted item-removal iterations when one or both of two possible results occurred: (1) the χ^2 difference tests showed no difference and/or (2) the AGFI did not increase. For the hedonic dimension, AGFI hit a plateau at five items (AGFI = .88); thus, we selected this five-item version of the hedonic scale (Table 1). For the utilitarian subscale, the χ^2 difference test and decreased AGFI also suggested a five-item scale (Table 1). Psychometrically, this ten-item (5 + 5) HED/UT scale performs better than the original version (χ^2 = 175.06 [d.f. = 34, $p < .0001$]; GFI = .95; NFI, NNFI, and CFI = .97). Coefficient alpha, composite reliability, and AVE (Fornell and Larcker 1981) exceed recommended standards (Table 1).

Second-order relationship. Thus far, we have analyzed the HED/UT scales separately; here, we test whether the two dimensions link to a higher-order construct (Bollen 1989) by fitting a second-order factor analysis model (Jöreskog and Sörbom 1996). The second-order factor analysis model supports our contention that hedonic and utilitarian product attitudes link to a common higher-order construct (χ^2 = 175.99 [d.f. = 35, $p < .0001$]; GFI = .95; NFI, NNFI, and CFI = .97). Standardized parameter estimates linking the higher-order construct to the utilitarian and hedonic constructs are identical (γ = .24) and significant (p = .032). Because the constructs were correlated (standardized ϕ = .37), we examined the discriminant validity between our HED/UT subscales. First, we fit an alternative model with all ten items loading on a single factor. If the two-factor model is superior, there should be a large difference in the χ^2 statistics associated with the two alternative models. In this case, the χ^2_Δ of 2802.88 (d.f. = 1) was significant with $p < .0001$. Second, Fornell and Larcker (1981) indicate that discriminant validity is achieved when AVE is greater than the squared correlation between the constructs. In this case, AVE estimates were greater than .70, and the squared correlation was .14. Finally, Anderson and Gerbing (1988) and Bagozzi, Yi, and Phillips (1991) recommend testing whether the correlation is less than one. We fit a two-factor model with the correlation set equal to one; a χ^2 difference test (χ^2_Δ = 4.81 [d.f. = 1, p = .0823]) suggests that the correlation between constructs is less than unity. Thus, the hedonic and utilitarian measures are distinct (i.e., they capture different information), but they are correlated through a relationship to a common higher-order construct.

Study 2B: Discriminant Validity

The literature lacks clarity regarding the relationship between involvement and the hedonic and utilitarian dimensions of attitude. For example, Mano and Oliver (1993) merge Batra and Ahtola's (1990) scale with Zaichowsky's (1985) 20-item involvement scale to model what they term utilitarian and affective dimensions. Kapferer and Laurent (1993) introduce a five-dimension conceptualization of involvement in which one dimension is labeled hedonic. We contend that the involvement construct is distinct from the hedonic and utilitarian dimensions of attitude. However, the research cited previously questions the discriminant validity between hedonic and utilitarian scales and measures of involvement. Thus, we tested discriminant validity using three published scales purporting to measure involvement. Scales collected in conjunction with Study 2A were the cognitive (PIICOG) and affective (PIIAFF) dimensions of Zaichkowsky's (1990) personal involvement inventory (PII), Traylor and Joseph's (1984) measure of involvement, and Kapferer and Laurent's (1993) five-dimension consumer involvement profile.

To determine the extent to which the HED/UT subscales captured information different from these three measures of involvement, we fit a series of single-factor and two-factor

[3]As Bollen (1989) details, the χ^2 test for CFAs has many weaknesses. Although throughout the article we present the χ^2 test, we rely primarily on GFIs in assessing model adequacy.

[4]As a reviewer noted, this design does introduce some dependency. However, we chose to maximize generality by having a large number of stimuli rated rather than maximize model fit by focusing on one or a few stimuli.

JMR3E.qxdII 7/3/03 9:51 AM Page 314

Table 2
DISCRIMINANT VALIDITY BETWEEN HED/UT AND MEASURES OF INVOLVEMENT

	Hedonic Dimension		*Utilitarian Dimension*	
	r	χ^{2a}_{Δ}	*r*	χ^{2a}_{Δ}
PII				
Cognitive	.48	1884.25	.78	1102.37
Affective	.82	289.68	.58	2956.52
CIP[b]				
Importance	.39	276.90	.38	278.85
Mispurchase	.16	493.05	.03	4349.85
Symbolism	.32	841.54	.12	928.51
Hedonic	.67	833.70	.27	1331.36
Interest	.57	320.95	.42	390.13
TJI[c]	.47	1987.47	.22	2460.93

[a]All χ^2 differences have 1 d.f. with $p < .0001$.
[b]Consumer involvement profile (Kapferer and Laurent 1993).
[c]Consumer involvement in products (Traylor and Joseph 1984).

CFA models in LISREL. For example, we fit a single-factor model with our hedonic items loading on the same factor as the five items from PIIAFF and compared the fit of that model with a similar CFA with our hedonic subscale loading on a different factor than did PIIAFF. If the two-factor model is superior, there should be a reduction in the χ^2 statistic relative to the single-factor model. In every case, the two-factor model provided a vastly superior fit (Table 2). Because HED-PIIAFF and UT-PIICOG were correlated (Table 2), we took the additional step of testing whether the construct correlations were less than unity (Anderson and Gerbing 1988; Bagozzi, Yi, and Phillips 1991). We fit a two-factor CFA model with the correlation between the constructs fixed at unity (Jöreskog and Sörbom 1996) and found that the construct correlation was less than one in both cases (HED: $\chi^2_{\Delta} = 354.13$ [d.f. = 1, $p < .0001$]; UT: $\chi^2_{\Delta} = 341.15$ [d.f. = 1, $p < .0001$]). Finally, we compared the AVE with the squared correlation between the constructs, and in both cases (HED: .71 > .67; UT: .79 > .61) AVE exceeded the squared correlation (Fornell and Larcker 1981). Taken together, this evidence suggests that when used to measure attitudes of product categories, HED/UT captures information different from the affective and cognitive dimensions of product category involvement.

Studies 3, 4, and 5: Criterion (Predictive) Validity

Study 3. Study 3 tests whether HED/UT discriminates among product categories expected a priori to differ along the hedonic and utilitarian dimensions. In selecting categories for the study, we used a three-step process. First, we identified products that had been used in prior literature (i.e., Batra and Ahtola 1990; Crowley, Spangenberg, and Hughes 1992). The product categories we identified were blue jeans, athletic shoes, automobiles, and vacation resorts; empirical evidence has shown that these product categories are high in both utilitarian and hedonic ratings. Second, because prior research has failed to identify adequately products that would be low in either the hedonic or the utilitarian ratings, we asked 129 students (who were given definitions of the hedonic and utilitarian dimensions) at two North American universities to identify one product category that would fall into each of three classifications: (1) high utilitarian/low hedonic value, (2) low utilitarian/high hedonic value, or (3) low utilitarian/low hedonic value. We then asked students to name five brands in each product category. We selected for further study the four categories in each of two hypothesized quadrants that received the most mentions: low hedonic/high utilitarian (disposable diapers, shoelaces, alkaline batteries, and paper clips) and high hedonic/low utilitarian (tobacco, beer, video games, and television sets). This elicitation produced no usable responses for the low hedonic/low utilitarian quadrant. Thus, as a third step, we relied on brainstorming to derive categories for this low/low quadrant, which includes plastic fruit, pet rocks, fake mustaches, and glass figurines. We collected evaluations of these 16 product categories from 567 students in four lab sessions at a large North American university. We randomly assigned subjects to rate 8 of the 16 product categories using HED/UT and Batra and Ahtola's (1990) scale. We fit a CFA in LISREL for each product category to check the performance of HED/UT. These results, which further substantiate the performance of the HED/UT scale, appear in Table 3 along with associated coefficient alphas.

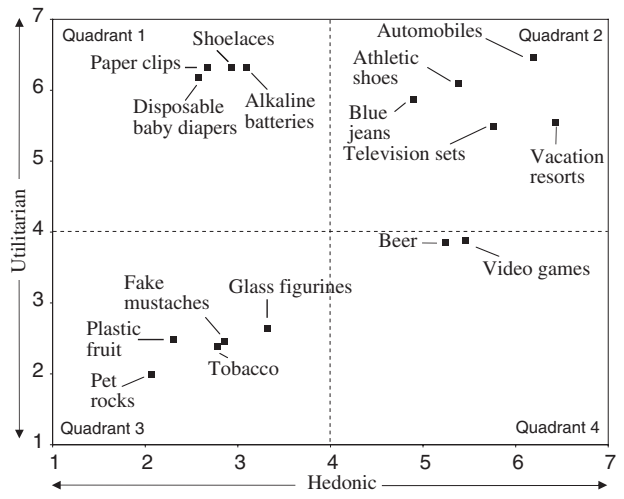
Table 3
TWO-FACTOR HEDONIC AND UTILITARIAN CFAs USING PRODUCT CATEGORIES

	n	χ^2	*d.f.*	*p-Value*	*GFI*	*NFI*	*NNFI*	*CFI*	$\alpha_{hedonic}$	$\alpha_{utilitarian}$
Pet rocks	379	230.82	34	.0000	.88	.93	.92	.94	.93	.94
Tobacco	341	96.02	34	.0000	.95	.96	.96	.97	.93	.85
Shoelaces	379	171.51	34	.0000	.92	.94	.93	.95	.85	.95
Blue jeans	377	176.37	34	.0000	.91	.90	.89	.92	.83	.89
Athletic shoes	380	259.76	34	.0000	.88	.89	.87	.90	.86	.90
Glass figurines	376	201.52	34	.0000	.90	.93	.92	.94	.93	.90
Video games	376	160.70	34	.0000	.92	.94	.94	.95	.94	.87
Alkaline batteries	377	166.09	34	.0000	.91	.93	.92	.94	.82	.90
Paper clips	233	75.01	34	.0001	.94	.94	.96	.97	.83	.85
Automobiles	238	81.09	34	.0000	.94	.92	.94	.95	.85	.82
Fake mustaches	226	99.23	34	.0000	.92	.93	.94	.95	.89	.88
Television sets	199	104.53	34	.0000	.90	.90	.90	.93	.91	.78
Plastic fruit	173	65.41	34	.0010	.93	.93	.95	.96	.88	.84
Beer/alcoholic beverages	144	100.52	34	.0000	.87	.89	.90	.93	.92	.85
Disposable baby diapers	181	58.42	34	.0057	.94	.94	.96	.97	.88	.84
Vacation resorts	179	58.84	34	.0052	.94	.93	.96	.97	.89	.82
Average					.92	.93	.93	.95	.88	.87

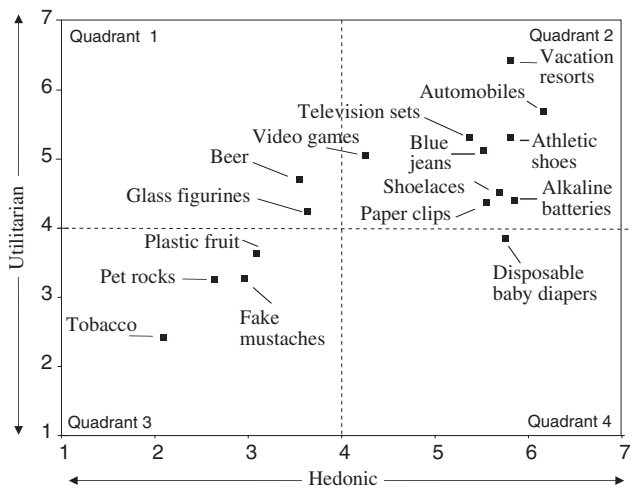
JMR3E.qxdII 7/3/03 9:51 AM Page 315

Figure 1
CLASSIFICATION OF PRODUCT CATEGORIES ON HEDONIC AND UTILITARIAN DIMENSIONS
(COMPARISON OF HED/UT AND BATRA AND AHTOLA [1990])

Product Categories Classified Using HED/UT



Product Categories Classified Using Batra and Ahtola (1990)



To assess predictive validity, we created an x–y plot, following Crowley, Spangenberg, and Hughes (1992), and identified quadrants by scale midpoints. In Figure 1, 14 of the 16 product categories plotted as hypothesized. The two misclassified items had been hypothesized to fall in Quadrant 4; tobacco actually plotted in the low utilitarian/low hedonic quadrant, and television sets received high ratings on the utilitarian dimension and the hypothesized hedonic dimension. Analysis of variance indicated that smokers and former smokers rated tobacco as significantly more hedonic ($F = 129.33$, $p < .001$) and utilitarian ($F = 53.43$, $p < .001$) than did nonsmokers; mean ratings for smokers therefore place tobacco squarely in the high hedonic/low utilitarian quadrant, as our elicitation suggests. Television sets' appearing in Quadrant 2, though contrary to the elicitation, is consistent with research that finds stereo systems high in both hedonic and utilitarian ratings (Crowley, Spangenberg, and Hughes 1992).

After reverse scoring, ratings of the 16 categories using Batra and Ahtola's (1990) scale (Figure 1) enabled comparison with our HED/UT. Of the 16 product categories, 8 are misclassified using the Batra and Ahtola scale, including 3 of the 4 low hedonic/high utilitarian categories falling in the high hedonic/high utilitarian quadrant. Psychometrically, Batra and Ahtola's scale was inferior to HED/UT; across all product categories, coefficient alpha averaged .80 and .77 for Batra and Ahtola's scales, respectively, compared with alpha of .88 and .87 for our HED/UT. The EFA of Batra and Ahtola's scale shows single-factor solutions for 5 of the categories. This psychometric evidence, the superior predictive ability of HED/UT, and critical evaluations by others (Crowley, Spangenberg, and Hughes 1992; Mano and Oliver 1993) further support our belief that HED/UT is superior to Batra and Ahtola's scale.

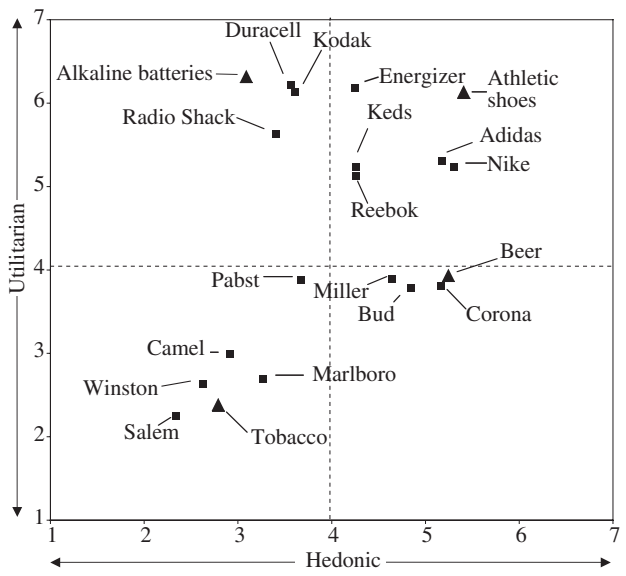
Study 4. In extending the predictive validity of HED/UT to brand names, we selected one product category from each quadrant in Figure 1: batteries, athletic shoes, beer, and tobacco. For each category, we compiled the four brands that received the most mentions in the Study 3 elicitation. We then asked 142 student subjects from a (different) North American university to each rate a set of four randomly assigned brand name products, one from each category. After eliminating nonrespondents, 500 observations were available for analysis. A multiple-groups CFA again confirmed the unidimensionality and reliability of HED/UT ($\chi^2 = 324.67$ [d.f. = 136, $p < .001$]; GFI = .90; NFI = .90; NNFI = .91; CFI = .93; and $\alpha = .93$ [hedonic] and .92 [utilitarian]).

Prediction of brand placement on the quadrant map is not straightforward because of the relationship between a product category and its associated brands. A person's evaluation of a product category is a summary construct of the brands in a category (e.g., Chaudhuri and Holbrook 2001) and may operate in several ways. For example, the product category evaluation may derive from an evaluation of the prototypical brand within the category, or the category evaluation may be an average evaluation of the brands within the category. Regardless of the underlying process, we expect variability in subjects' ratings of brands within a product category. For example, if a brand adopts an experiential positioning strategy (Park, Jaworski, and MacInnis 1986), it may be positioned higher than competing brands on the hedonic dimension. Thus, it is possible that brands in a single product category may fall in different quadrants. Figure 2 shows specific brand locations relative to respective product category positions.

In confirmation of our expectation, two product categories had brands that appeared in two quadrants. For batteries, Kodak and Duracell are located close together with nearly the same utilitarian rating estimate for the overall category; relative to the reference, however, Energizer is given nearly equivalent utilitarian ratings but higher hedonic ratings, and Radio Shack is given nearly equivalent hedonic ratings but lower utilitarian ratings. Across these four product categories, brand attitudes appear to be associated with

JMR3E.qxdII 7/3/03 9:51 AM Page 316

Figure 2
CLASSIFICATION OF BRAND NAME PRODUCTS ON HEDONIC AND UTILITARIAN DIMENSIONS USING HED/UT



Notes: Triangles indicate average product category ratings (from Figure 1).

the attitude toward the product category, but the number of brands included is insufficient to make strong generalizations in this regard. A notable, if unanticipated, finding is the tendency of the brands to vary on the hedonic dimension more so than on the utilitarian dimension. This is apparent in the beer and athletic shoe categories and in the battery category, with the exception of Radio Shack (utilitarian attitudes could result from Radio Shack's status as the only store brand in the study). Tobacco brands vary on the utilitarian dimension, but as in Study 2B, smokers drive this difference.

Study 5. Although Studies 3 and 4 demonstrate predictive validity, perhaps a more stringent criterion test of a measure is its usefulness in marketing research. Thus, we measured subjects' hedonic and utilitarian attitudes toward a fictional brand of soap (Cue, chosen by pretest [n = 67] as relevant to students) after we manipulated brand positioning. We created a product concept description and two alternative advertising concepts with either functional or experiential positioning. Functional positioning statements stress a brand's ability to solve consumption-related problems, whereas experiential positioning statements stress a brand's effects on sensory satisfaction (Park, Jaworski, and MacInnis 1986). We exposed pretest subjects to five statements for functional positioning and five for experiential positioning. For bath soap, subjects perceived "strong" and "exotic" as functional and experiential attributes, respectively, and thus we chose these words for the positioning statement manipulations. Comparative advertising research indicates that direct comparative claims are superior for positioning and are most effective when the referent brand is a market leader (Dröge and Darmon 1987). Thus, we used a leading national brand of soap (Ivory) as the referent in direct comparative advertisements. In the functional positioning condition, the advertising concept contained a picture of a cheetah and a mock label with the brand's name and product claims such as "antibacterial" and "retains skin moisture." The word "strong" appeared under the picture, and the phrase "As Strong as Ivory" appeared under the mock label. The stimuli for the experiential positioning condition were identical, except that a picture of a waterfall replaced the picture of the cheetah and "strong" was replaced with "exotic" wherever it appeared. The HED/UT should measure respective differences ensuing from the experiential and functional positioning statements. The elaboration likelihood model (ELM) suggests that product category involvement moderates brand ratings (Petty, Cacioppo, and Schumann 1983). Thus, we measured involvement with the PII; a median split delineated high and low involvement groups. Consequently, we employed a 2 (involvement: high or low) × 2 (positioning statement: functional or experiential) design.

We randomly assigned students at a large North American university (n = 164) who had not participated in our previous studies to lab sessions in which they were exposed to one of the two manipulated conditions. Using Hotelling's T, a multivariate analysis of variance showed significant results for positioning statement (F = 8.26 [d.f. = 2158, p = .0003]) and product category involvement (F = 5.36 [d.f. = 2158, p = .0056]). In support of Dröge's (1989) finding, interaction effects were not significant (F = 1.08 [d.f. = 2158, p = .3438]). Investigation of the between-subjects effects indicated that the model was significant for the hedonic dimension (F = 8.60 [d.f. = 3159, $p < .0001$]) but not the utilitarian dimension (F = .92 [d.f. = 3159, $p < .4322$]). In the hedonic model, the main effect for positioning was significant (F = 15.80 [d.f. = 1159, $p < .0001$]), as was the main effect for involvement (F = 10.77 [d.f. = 1159, p = .0013]). As predicted, the mean hedonic attitude was higher in the experiential positioning condition for both involvement groups (low F(1, 75) = 5.49 [p = .0219]; high F(1, 75) = 10.98 [p = .0009]). Failure to observe differences on the utilitarian dimension suggests that subject responses to brand positioning advertisements are perhaps more complex than the literature has shown. Our Study 4 data suggest that brands do not vary much on the utilitarian dimension. One suggestion is that functional positioning statements may not be effective for brand differentiation within homogeneous product categories. Study 5 shows that at least on the hedonic dimension, the HED/UT scale is capable of capturing differences in subjects' brand attitudes in response to advertising positioning statements.

Study 6: Nomological Validity

Attitude toward the brand is a focal construct in attitude-change research (e.g., Brown and Stayman 1992; Homer 1990; MacKenzie, Lutz, and Belch 1986; Mittal 1990). Attitude-change thinking is firmly rooted in the ELM (Petty, Cacioppo, and Schumann 1983). The ELM proposes that attitude change can occur in two ways. The central route describes the process of attitude change when message-relevant thinking (i.e., elaboration) is high, and the peripheral route describes attitude change when message-relevant thinking is low (Petty, Cacioppo, and Schumann 1983). For example, when the product is personally relevant, the consumer likely spends more time elaborating the message (Petty, Cacioppo, and Schumann 1983). In this stream of research, the central route to persuasion is modeled as the

JMR3E.qxdII 7/3/03 9:51 AM Page 317

chain of effects from advertising exposure to purchase intention (MacKenzie, Lutz, and Belch 1986), and cognitive elaboration directly antecedes the mediating attitude toward the brand (e.g., Brown and Stayman 1992; Goldsmith, Lafferty, and Newell 2000) in predicting purchase intent (e.g., Miniard, Bhatla, and Rose 1990).

The extent to which a measure defines a construct depends on how well the measure fits into a network of expected relationships called a "nomological network" (Nunnally and Bernstein 1994). The central route is such a network. Inferences about nomological validity will follow if our HED/UT scales have significant relationships with purchase intent and mediate the relationship between cognitive elaboration and purchase intent (Churchill 1995). If HED/UT is an equal or superior substitute for the A_b measure in the central route processing model, we have evidence of nomological validity because HED/UT would replace its higher-order relative in a well-established nomological network. Thus, our nomological validity test compares a central route processing model using the hedonic and utilitarian constructs, as measured by our HED/UT scales, with one including A_b (Nunnally and Bernstein 1994; Olney, Holbrook, and Batra 1991).

We used the central route processing models shown in Figure 3, which are limited versions of the dominant conceptualization in brand attitude research in that they focus solely on the central route to persuasion. In Model 1, we view A_b as directly antecedent to purchase intentions. It is well established, both theoretically and empirically, that the amount of elaboration is positively associated with involvement (Brown and Stayman 1992; Homer 1990; Lord, Lee, and Sauer 1995; Miniard, Bhatla, and Rose 1990; Mittal 1990; Petty, Cacioppo, and Schumann 1983). Thus, with central route persuasion, involvement can be a proxy for cognitive elaboration (Petty, Cacioppo, and Schumann 1983). As MacKenzie and Lutz (1989) note, measures of cognitions may be weak as a result of low reliability; thus, we measure product category involvement as a proxy for cognitive elaboration and propose in Model 1 that attitude toward the brand (A_b) depends on product category involvement. We used the two-dimensional view of involvement (Zaichkowsky 1990) in which affective involvement is associated with the personal relevance of the product as a result of emotional attachment, and cognitive involvement arises when personal relevance is based in logic.

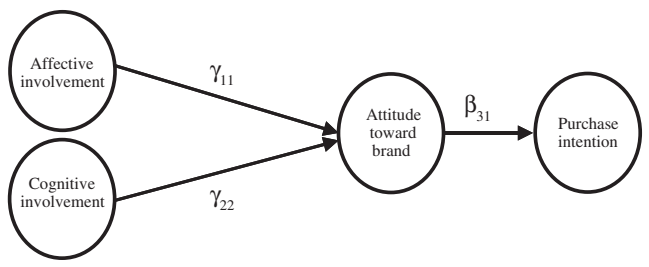
Consistent with prior research regarding the experiential view of consumption (Hirschman and Holbrook 1982) and our arguments and empirical evidence, we conceptualize the hedonic and utilitarian constructs as dimensions of brand attitude. Thus, we propose Model 2, in which we substitute the hedonic and utilitarian constructs for attitude toward the brand in the central route processing model. In this competing model, consistent with Study 2B, affective involvement predicts the hedonic dimension, and cognitive involvement predicts the utilitarian dimension.[5]

Method. To compare the proposed central route processing models in Figure 3, we selected four brand name products that represented each of the four quadrants in Figure 1: Duracell batteries, RCA compact disc (CD) players, Hershey's chocolate, and Winston cigarettes (chosen either from the elicitation procedure described in Study 3 or from the pretest).[6] In controlled lab sessions, student subjects at a large North American university (n = 264) first completed PII measures (Zaichkowsky 1990) for all four stimuli. After participating in a 20-minute distraction task, we exposed subjects to advertisements extracted from published shopping circulars. The advertisements had a picture of the brand and its price without any peripheral execution cues about the brands or references to the offering merchants.[7] Because a picture of the product and price are rational message arguments (Goldsmith, Lafferty, and Newell 2000), these advertisements should be persuasive for those who are highly involved but less so for those who are uninvolved. Thus, persuasion in this study should occur only along the central route. Subsequently, subjects completed a booklet for a single product with instructions on the outside cover followed by HED/UT, a measure of A_b (MacKenzie, Lutz, and Belch

Figure 3
CENTRAL ROUTE PROCESSING MODEL FOR NOMOLOGICAL VALIDITY COMPARISON

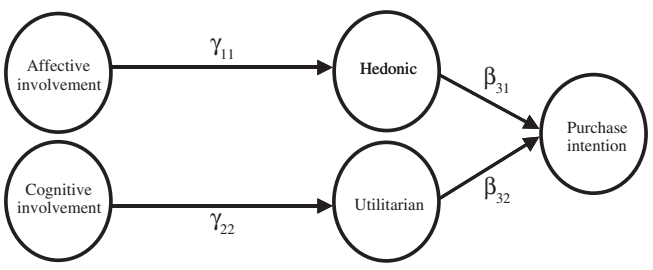
Model 1

Central Route Processing Model with One-Dimensional Measure of Attitude Toward the Brand



Model 2

Central Route Processing Model with Hedonic and Utilitarian Dimensions of Attitude



[5]We are indebted to the reviewers for suggesting this approach.

[6]Because chocolate candy (high hedonic/low utilitarian) and CD players (high hedonic/high utilitarian) had not yet been measured with our ten-item HED/UT, we conducted a pretest with new subjects to confirm that they would fall in the hypothesized quadrants. Student subjects rated chocolate candy (n = 64) and CD players (n = 45); results confirmed that these products fall cleanly into the expected quadrants.

[7]As a reviewer noted, involvement is typically viewed as a moderator in the ELM. As befits a nomological validation study, however, we do not seek to make strong inferences about the attitude formation process. Thus, we do not manipulate advertising stimuli or use image advertisements. It bears emphasis that more complete models of attitude change recognize the peripheral route to persuasion.

JMR3E.qxdII 7/3/03 9:51 AM Page 318

1986; Miniard, Bhatla, and Rose 1990), and a measure of purchase intentions (Dodds, Monroe, and Grewal 1991; Netemeyer and Bearden 1992).

Results. The psychometric properties of HED/UT compared favorably with results previously reported about standard tests of reliability and unidimensionality.[8] We fit the data to the structural models in Figure 3 using a multiple-groups design in LISREL 8. Subsequently, we refit the model with all structural parameters constrained to equality across groups. A χ^2 difference test ($\chi^2_\Delta = 123.46$ [$d.f._\Delta = 15$, $p < .0001$]) indicates that the structural parameters (Table 4) are not equal across groups; thus, the model is moderated by the quadrant from which the brand was chosen (i.e., by product). A comparison of Model 1 with Model 2 by a χ^2 difference test is not appropriate because the models are nonnested. Consequently, we compared the two models on several criteria (Morgan and Hunt 1994). First, we examined the overall fit of the two models. Second, we compared the models' ability to explain variance in purchase intentions using squared multiple correlations. Third, we compared the two models using the Bayesian cross-validation likelihood (BCVL) procedure that Rust, Lee, and Valente (1995) recommend for nonnested models. The BCVL assesses the likelihood that Model 2 emerges as the better model given the prior probability that Model 1 is correct. Although the prior probability of Model 1 is unknown, a range of priors is used in sensitivity analysis (Rust, Lee, and Valente 1995).

[8]An appendix containing factor loadings, AVE, and reliability estimates (Fornell and Larcker 1981) for all measures used in Study 6 is available on request from the first author.

As the fit indexes in Table 4 indicate, both Model 1 and Model 2 are good fits given the data, and thus both should be considered qualified models (Rust, Lee, and Valente 1995). Although Model 2 has a slight edge, any conclusion at this point is tentative because of the increase in degrees of freedom. The R^2 estimates in Table 4 show mixed results. For Duracell, Model 1 has a higher explained variance, and Model 2 dominates for Hershey's and Winston. Model 2 has a slight edge in explaining purchase intentions for the RCA-brand CD player. Finally, BCVL results, at three choices of prior probabilities for Model 1 (.50, .90, .95), show a clear preference for Model 2 (Table 4). Interpretation of these results leads to the conclusion that even if we were 95% confident before the study that the A_b model (Model 1) was the correct model, there is an 80% probability that the HED/UT model (Model 2) would prove superior when confronted with data.

GENERAL DISCUSSION

A main goal of this research was to establish reliable, valid, generalizable, and useful measures of the hedonic and utilitarian dimensions of overall brand/product attitudes. This goal has largely been realized. The ten-item HED/UT scale demonstrated solid performance in several psychometric tests and in multiple tests of criterion and discriminant validity. We replicated reliability and validity with several separate samples at separate geographic locations and across a wide variety of stimuli. Furthermore, HED/UT successfully measured attitudes toward both product categories and brands in those product categories, though more research is needed before strong generalizations in this area are possi-

Table 4

NOMOLOGICAL VALIDITY MODEL PARAMETER ESTIMATES AND MODEL FIT COMPARISON

Standardized Parameter Estimates

	Model 1[a]				*Model 2*[b]			
	Duracell Batteries	*RCA CD Player*	*Hershey's Chocolate*	*Winston Cigarettes*	*Duracell Batteries*	*RCA CD Player*	*Hershey's Chocolate*	*Winston Cigarettes*
γ_{11}	.10*	.14	.48*	.18	.49*	.26*	.88*	.52*
γ_{22}	–.14	.17*	.44*	.52*	–.09	.19*	.89*	.78*
β_{31}	.60*	.88*	.75*	.46*	.09	.55*	.78*	.70*
β_{32}	—	—	—	—	.74*	.36*	.26*	.31*
R^2	.35	.50	.62	.44	.15	.53	.71	.66

Model Fit Comparison

				Independence Model					
	χ^2	*d.f.*	*p*	χ^2	*d.f.*	*GFI*	*NFI*	*NNFI*	*CFI*
Model 1[a]	32.49	8	.0001	580.95	24	.90	.94	.85	.95
Model 2[b]	17.81	16	.3200	858.01	40	.97	.98	.99	.99

BCVL Posterior Probabilities

P(Model 1)[c]	*.50*	*.90*	*.95*
Model 1[a]	.01	.11	.20
Model 2[b]	.99	.89	.80

*Parameter estimates significant at $\alpha = .01$.
[a]Corresponds to Model 1 in Figure 3.
[b]Corresponds to Model 2 in Figure 3.
[c]The prior probability that Model 1 is the correct model.

JMR3E.qxdII 7/3/03 9:51 AM Page 319

ble. Notably, an additional benefit of our research emerged. In reducing the psychometrically sound but otherwise too long scale resulting from our initial development efforts, we developed a process for paring down a set of items to arrive at a more parsimonious, manageable scale. This item-reduction process should be of value to future measurement development efforts.

We demonstrated the value to marketing researchers of adopting a two-dimensional (hedonic and utilitarian) view of product attitude by consumers in several ways and across multiple studies. First, we demonstrated discriminant validity between hedonic and utilitarian attitudes and product category involvement. Second, we showed hedonic and utilitarian benefits to be important dimensions of overall attitude toward the brand. Second-order factor analysis indicated that the dimensions of HED/UT are related to a higher-order construct, and the nomological validity test supported our contention that this construct was brand attitude. Third, predictive validity studies demonstrated that HED/UT could be useful for assessing brand differences and advertising effectiveness. Finally, echoing findings of previous research, we linked the hedonic and utilitarian dimensions of attitude, such as attitude toward the brand, with purchase intentions. In this vein, our nomological validity study demonstrated that using separate hedonic and utilitarian attitudinal dimensions produces a stronger model than using a one-dimensional measure of brand attitude. Thus, our HED/UT scale provides marketing researchers with more worthwhile and complete measures of a complex phenomenon that has traditionally been measured with a single dimension: A_b.

CONCLUSION

The studies reported herein support the contention that complex processes are at work in the formation of consumers' attitudes toward brands/products and their development of purchase intentions. Specifically, it appears that the hedonic and utilitarian constructs are separate and important dimensions of attitude toward products and brands. Limiting the scope of our nomological model to central route processing provided evidence of validity while somewhat constraining the generalizations that can be made. We intentionally restricted our development of the model in Study 6; it was designed to test our HED/UT dimensions versus A_b in a small nomological network. Further research should examine incorporating the hedonic and utilitarian dimensions of attitude in more complex models with constructs such as attitude toward the ad (McKenzie, Lutz, and Belch 1986) and ad skepticism (Obermiller and Spangenberg 1998).

Another concern is the failure of the functional positioning statement to generate significant differences on the utilitarian dimension in Study 5. It is possible that the comparative nature of the positioning statement erased any perception of utilitarian differences, but we believe this is unlikely. The experiential positioning statement also included a comparative claim, but significant differences resulted on the hedonic scale. If the comparative claim erased differences on the utilitarian scale, it should erase differences on the hedonic scale as well. Regarding the interaction between positioning statement and involvement, the nonsignificant results are in concert with previous findings in comparative advertising research (Dröge 1989). Nonsignificant effects for level of involvement on the utilitarian scale can be interpreted to mean that customers understand the utilitarian properties of bar soap, regardless of their level of involvement. Thus, when products are highly functional, involvement may play less of a role in consumers' evaluations of brands. Accordingly, the product used could be (and is, in our view) responsible for the nonsignificant findings. Additional research dedicated to sorting out the roles of comparative statements, involvement, and product/brand choice would be valued.

The development and introduction of HED/UT—a reliable, valid, parsimonious, and generalizable scale for measuring two important underlying dimensions of consumer attitude—should encourage and facilitate further testing of postulations in this stream of research. The measure should be of value to researchers, because greater variance is accounted for by HED/UT than by traditional one-dimensional brand attitude measures. By conceptualizing consumer attitudes as comprising two dimensions and by using HED/UT to capture these dimensions, marketing researchers can develop a richer understanding of critical marketing outcomes.

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JMR3E.qxdII 7/3/03 9:51 AM Page 320

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